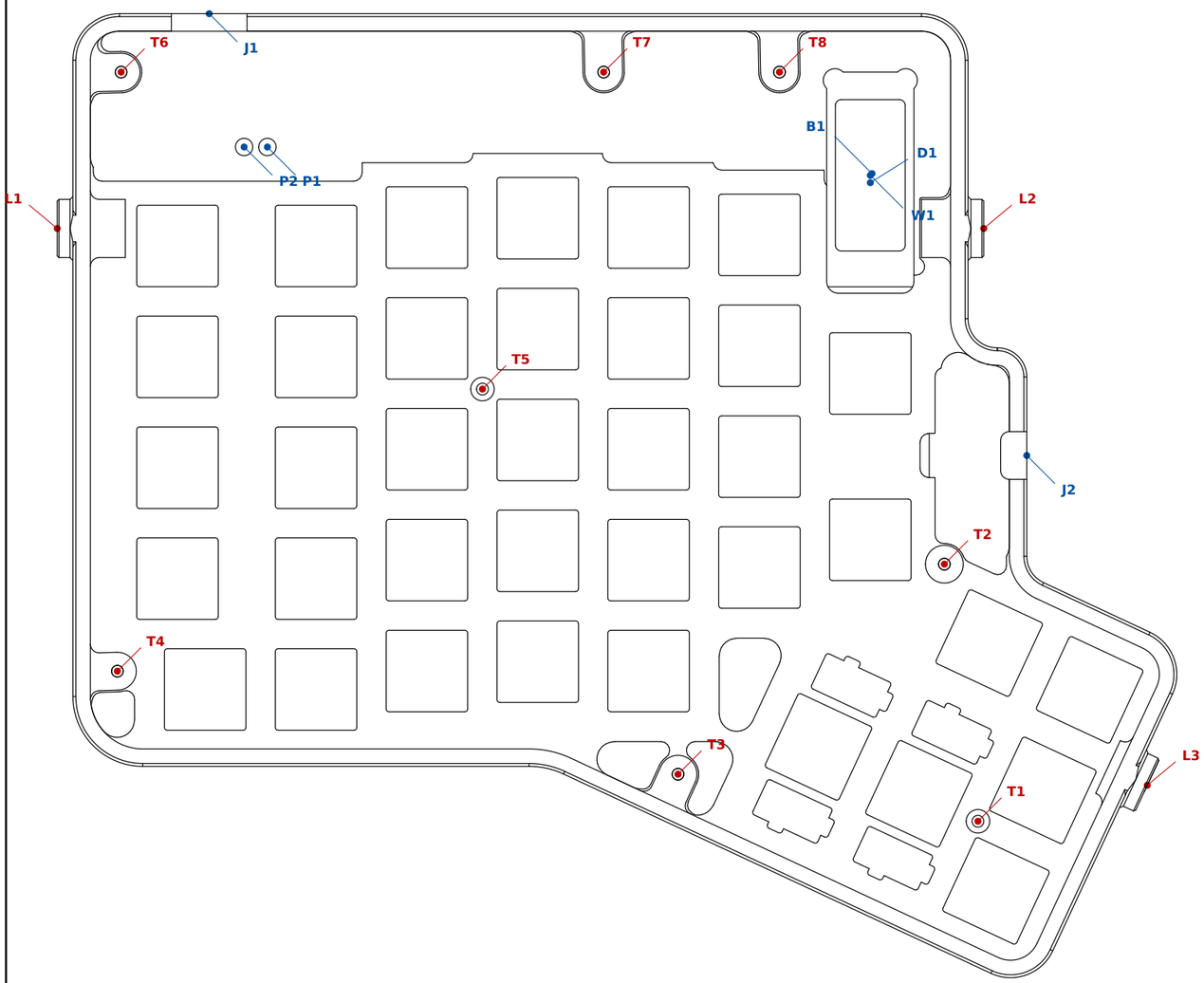
VIEW FROM BELOW (−Z), parting face up — scale 1:1



ID	Thread	Hand	Cls	Tap drill	Drill depth	Min full thr.	Screw	X	Y	Z	Note
T1	M2.5x0.45	RH	6H	Ø2.05	thru	5.0	M2.5x12 CSK	-76.10	-65.84	8.6	THRU to key-well floor, exit Ø2.9 x 90°, note 5
T2	M2.5x0.45	RH	6H	Ø2.05	6.0	4.5	M2.5x12 CSK	-70.34	-21.67	8.6	blind
T3	M2.5x0.45	RH	6H	Ø2.05	6.0	4.5	M2.5x12 CSK	-24.56	-57.76	8.6	blind
T4	M2.5x0.45	RH	6H	Ø2.05	6.0	4.5	M2.5x12 CSK	71.78	-40.06	8.6	blind
T5	M2.5x0.45	RH	6H	Ø2.05	thru	5.0	M2.5x12 CSK	9.04	8.41	8.6	THRU to key-well floor, exit Ø2.9 x 90°, note 5
T6	M2.5x0.45	RH	6H	Ø2.05	6.0	4.5	M2.5x12 CSK	71.14	62.87	8.6	blind
T7	M2.5x0.45	RH	6H	Ø2.05	6.0	4.5	M2.5x12 CSK	-11.79	62.87	8.6	blind
T8	M2.5x0.45	RH	6H	Ø2.05	6.0	4.5	M2.5x12 CSK	-42.00	62.87	8.6	blind
L1	M3x0.5	RH	6H	Ø2.5	5.5	4.0	M3 leg thumbscrew	82.12	36.00	12.0	blind, in the flat leg boss, see sheet 2
L2	M3x0.5	RH	6H	Ø2.5	7.5	6.0	M3 leg thumbscrew	-77.10	36.00	12.0	blind, in the flat leg boss, see sheet 2
L3	M3x0.5	RH	6H	Ø2.5	5.5	4.0	M3 leg thumbscrew	-105.21	-59.66	12.0	blind, in the flat leg boss, see sheet 2

X/Y/Z = hole start point in STEP coordinates. T holes: drilled +Z from the PCB face of the boss. L holes: drilled into the part, normal to the wall, from the boss face.
Drill depth = full-diameter depth; the 118° drill point is extra (modelled). Blind holes keep at least D/2 unthreaded at their bottom. T1 and T5: drilled and tapped through.

ID	Feature (no thread)	X	Y
W1	display window 12 x 26 R1.0 THRU the bezel skin, 0.5 x 45° chamfer on the top edge	-57.61	45.13
D1	display pocket 15.0 x 38.0, Z 8.6-18.6 (skin 1.0), 11.6 deep from the parting face: R2.0 dog-bones at the 2 back corners (keep), front end open into the key well	-57.61	43.87
B1	raised display bezel 18.42 x 38.90, top Z 19.6 (2.0 above the top face), top edges 0.5 x 45°, R2.0 free corners; flush with the key-well walls on its well sides	-57.90	45.43
J1	USB-C slot 13.0 +0.2/0 wide THRU the back wall, open to the parting face, top at Z 14.2 (it continues 0.8 into the bottom part)	56.00	72.90
J2	TRRS slot 8.2 +0.2/0 wide THRU the inner bump wall, open to the parting face, top at Z 15.2; 1.5 past the wall as a step in the pocket ceiling, R1.5 end corners	-84.50	-3.00
P1	Ø3.0 THRU top skin (SW_BOOT pin-hole)	46.00	50.00
P2	Ø3.0 THRU top skin (SW_RST pin-hole)	50.00	50.00

GENERAL NOTES

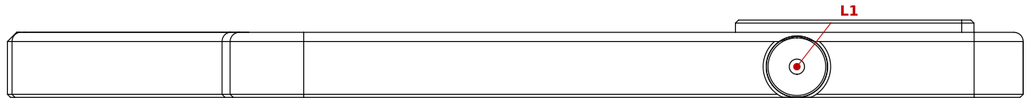
1. Geometry per worxboard_right_top.step. Dimensions not shown are taken from the STEP model. Overall size 192.4 x 165.6 x 12.6 (reference).
2. Material: Aluminium 6061-T6 (7075-T6 acceptable).
3. THREADS: 8x M2.5x0.45-6H and 3x M3x0.5-6H, ISO metric coarse, ALL RIGHT-HAND (clockwise to tighten), blind except T1 and T5 (through). Holes are modelled at tap-drill diameter in the STEP; tap them per the table to at least the minimum full-thread depth. The screw column names the screw each hole takes; its tip stops at least 0.3 short of the end of the full thread.
4. The left-hand part is a mirror image of this part. Its threads are also RIGHT-HAND.
5. T1 and T5 are under the key well: drilled and tapped THROUGH the boss and the plate (5.0 mm) and they exit in the cosmetic key-well floor, between two keycaps. Exit chamfer Ø2.9 x 90° as modelled, no burr, no break-out round the hole.
6. Switch cutouts (37x): 14.00 +0.05/-0 square, corner R0.5 max, plus 4 stabiliser cutouts. Plate thickness 1.50 ±0.05. Plate flatness 0.2 over the part in the restrained state (screwed to the bottom part).
7. FIT-CRITICAL, tighter than the general tolerance: T1-T8 positions ±0.1 hole to hole (they match C1-C8 of the bottom part and the PCB's holes). Cavity wall (inside of the perimeter wall, parting face to plate): +0.2/0 from the T1-T8 pattern, never inwards (the PCB clears it by 0.3). Boss faces (Z 8.6) to plate top (Z 13.6): 5.00 ±0.10. Parting face (Z 7.0) to boss faces: 1.60 +0.10/0.
8. Dimensions and tolerances apply AFTER finishing. Anodizing rack / contact marks on inside (non-cosmetic) faces only.
9. Internal corners are modelled at the minimum radius for their depth (R = depth/10 + 0.5; past 10 mm the Min R of the tool that reaches): key well R1.0 (4.0 deep), relief pockets and wall fillets R1.5 (to 7.6 deep from the parting face), pocket D1 R2.0 (11.6 deep, dog-bones at its back corners). Only the switch and stabiliser cutouts are R0.5 (Ø1 tool, through the 1.5 plate at the bottom of the open key well).
10. Display bezel B1 stands 2.0 mm above the top face (part height 12.6). Pocket D1 runs from the parting side up to 1.0 mm under the bezel top; window W1 opens it. Slots J1 (USB-C) and J2 (TRRS) are open to the parting face and cut from that side through the whole wall (see table): no side operation.
11. Leg bosses L1-L3: plain Ø10 bosses with a FLAT face (no teeth), see sheet 2.
12. Cosmetic surfaces: top face, display bezel, key well, outer walls, leg bosses: no tool marks, scratches or clamp marks. Inside (cavity) surfaces are non-cosmetic.
13. Break sharp edges 0.1-0.2 mm, except switch-cutout edges (0.1 max).
14. Finish: Fine bead blast + anodize Type II, colour per order. Mask threads during anodizing, or re-tap after anodizing.

worxboard CNC aluminium case

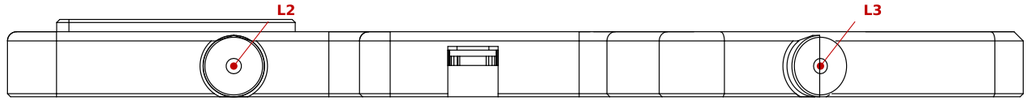
worxboard_right_top — https://github.com/w0rxbend/worxboard

Drawing no.: WXB-TOP-R	Sheet 1 of 2
Material: Aluminium 6061-T6 (7075-T6 acceptable)	Scale 1:1
Finish: Fine bead blast + anodize Type II, colour per order	Units: mm
Revision A5 Date 2026-11-12	Do not scale drawing
General tolerance ISO 2768-mK unless noted. Threads ISO metric, RIGHT-HAND, 6H.	
The STEP file governs geometry; this drawing governs threads, tolerances, finish.	

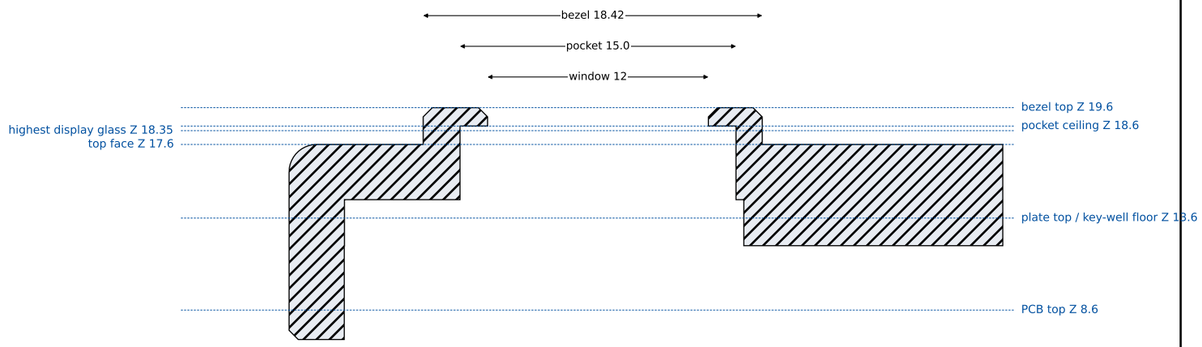
VIEW FROM +X — scale 1:1



VIEW FROM −X — scale 1:1



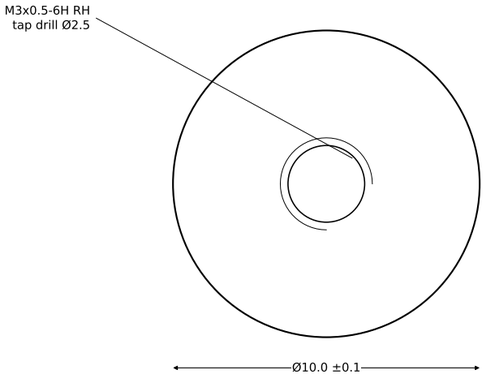
DETAIL C — SECTION THROUGH THE DISPLAY BEZEL at the window centre (X-Z, real section), scale 3:1

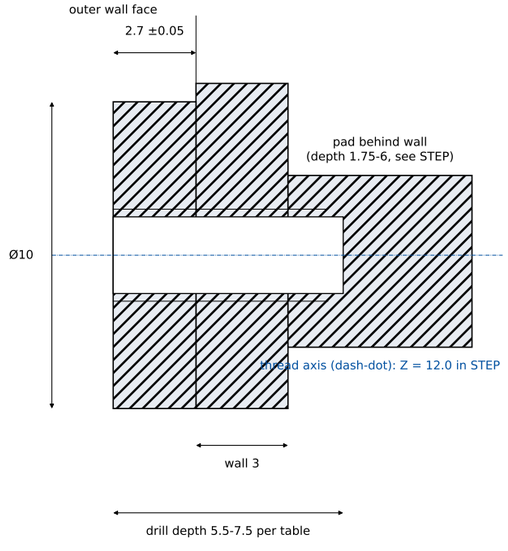


bezel skin 1.0; top edges 0.5 x 45°; window edge 0.5 x 45°. The bezel side wall on the key-well side is flush with the well wall.

SECTION B-B through boss axis, scale 5:1 (schematic)

DETAIL A — LEG BOSS L1-L3, face view, scale 5:1 (FLAT FACE, NO TEETH)





Leg boss specification	
Boss	Ø10.0 ±0.1, 2.7 ±0.05 proud of the outer wall face
Bearing face	FLAT, no teeth. Flatness 0.05, perpendicular to the thread axis within 0.05. Ra 1.6 max
Function	the leg hub is clamped against this face by the M3 spring thumbscrew (friction)
Thread	M3x0.5-6H RIGHT-HAND, blind, concentric with boss within Ø0.05; tap drill Ø2.5; depths per the table below
Quantity	3 bosses per part (L1-L3), identical
Machining hint	side op (4th axis, right-angle head or a setup per wall): face the boss, profile Ø10, drill and tap M3
Mating part	custom leg worxboard_leg (drawing WXB-LEG, flat hub face) + M3 thumbscrew

Boss	X face	Y face	Z	Thread direction (into part)	Drill depth	Min full thr.	Screw reach max
L1	82.12	36.00	12.0	into part (-1.000, -0.000, 0)	5.5	4.0	3.7
L2	-77.10	36.00	12.0	into part (+1.000, -0.000, 0)	7.5	6.0	5.7
L3	-105.21	-59.66	12.0	into part (+0.906, +0.423, 0)	5.5	4.0	3.7

X/Y/Z: centre of the boss face (tooth tip plane) in STEP coordinates. Depths from the boss face; the drill point is extra. >= 1.5 (D/2) stays unthreaded.

worxboard CNC aluminium case

worxboard_right_top — https://github.com/w0rxbend/worxboard

Drawing no.: WXB-TOP-R	Sheet 2 of 2
Material: Aluminium 6061-T6 (7075-T6 acceptable) Finish: Fine bead blast + anodize Type II, colour per order	Scale 1:1, 3:1, 5:1 Units: mm
Revision A5 Date 2026-11-12 General tolerance ISO 2768-mK unless noted. Threads ISO metric, RIGHT-HAND, 6H. The STEP file governs geometry; this drawing governs threads, tolerances, finish.	Do not scale drawing